Customer Segmentation & Prediction

Week 11 Internship Project — Business Recommendations

Dataset contains 500 customer records and 9 columns. Overall observed churn rate is 10.6%. Recommendations below are hypotheses to test, not guaranteed business outcomes.

Methodology

CustomerID and Churn were excluded from clustering features. Numeric features were median-imputed and standardized; categorical features were imputed and one-hot encoded. K-Means with three clusters was used for the main segment analysis. Hierarchical clustering and DBSCAN were also compared. A Random Forest model was trained per segment only where sample size and class diversity permitted; GridSearchCV tuned model parameters.

Segment profile summary

Segment	Customers	Share	Avg tenure	Avg monthly	Churn rate
Long-Tenure Value	114	22.8%	39.5	97.8	8.8%
Established High-Value	249	49.8%	35.7	108.1	10.0%
Newer High-Charge	137	27.4%	35.5	136.8	13.1%

Segment-specific actions

Long-Tenure Value — 114 customers; average tenure 39.5; average monthly charges 97.8; observed churn 8.8%.

Protect loyalty: reinforce value, recommend relevant add-ons, and use periodic satisfaction checks rather than blanket discounts.

Established High-Value — 249 customers; average tenure 35.7; average monthly charges 108.1; observed churn 10.0%.

Prioritize retention: proactive outreach, service check-ins, and a time-limited loyalty offer; measure churn reduction against a control group. Test premium support/value bundles; avoid discounts without margin analysis.

Newer High-Charge — 137 customers; average tenure 35.5; average monthly charges 136.8; observed churn 13.1%.

Prioritize retention: proactive outreach, service check-ins, and a time-limited loyalty offer; measure churn reduction against a control group. Add an early-life onboarding journey during the first 90 days. Test premium support/value bundles; avoid discounts without margin analysis.

Model evaluation

The accompanying model_evaluation_results.csv reports test-set accuracy, precision, recall, F1 and ROC-AUC for each segment where evaluation was feasible. Blank metrics mean the segment did not meet the minimum sample/class-diversity checks; they are not zero scores.

Business impact measurement plan

• Run retention offers as controlled experiments, comparing churn and contribution margin with a holdout group.

• Track segment-level churn, retention-offer acceptance, average revenue per user, and support contacts monthly.

• Estimate impact as eligible customers × measured incremental retention lift × expected contribution margin; do not use assumed lift as a reported result.

• Review fairness, privacy, model drift, and performance before deploying predictions to customer-facing workflows.